

Fundamentals of Linear Control:

A Concise Approach

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Chapter 7: Frequency Domain

linearcontrol.info/fundamentals

7.1 Bode Plots

Goal

Easily sketch the frequency response of rational transfer-function models

Setup

Given

$$G(s) = \frac{\beta (s - z_1)^{\ell_1} (s - z_2)^{\ell_2} \cdots (s - z_r)^{\ell_r}}{s^{k_0} (s - p_1)^{k_1} (s - p_2)^{k_2} \cdots (s - p_t)^{k_t}}$$

rewrite

$$G(s) = \frac{\tilde{\beta} (1 + \sigma_1 s)^{\ell_1} (1 + \sigma_2 s)^{\ell_2} \cdots (1 + \sigma_r s)^{\ell_r}}{s^{k_0} (1 + \tau_1 s)^{k_1} (1 + \tau_2 s)^{k_2} \cdots (1 + \tau_t s)^{k_t}}$$

where

$$\sigma_j = -z_j^{-1}, \quad \tau_i = -p_i^{-1}, \quad \tilde{\beta} = (-1)^{m-n} \frac{\beta z_1^{\ell_1} \cdots z_r^{\ell_r}}{p_1^{k_1} \cdots p_t^{k_t}}$$

Setup

Normalized transfer-function

$$G(s) = \frac{\tilde{\beta} (1 + \sigma_1 s)^{\ell_1} (1 + \sigma_2 s)^{\ell_2} \cdots (1 + \sigma_r s)^{\ell_r}}{s^{k_0} (1 + \tau_1 s)^{k_1} (1 + \tau_2 s)^{k_2} \cdots (1 + \tau_t s)^{k_t}}$$

Magnitude

$$\begin{aligned} 20 \log_{10} |G(j\omega)| = & 20 \log_{10} |\tilde{\beta}| - 20 k_0 \log_{10} |j\omega| + \\ & 20 \ell_1 \log_{10} |1 + j\sigma_1 \omega| + \cdots + 20 \ell_r \log_{10} |1 + j\sigma_r \omega| - \\ & 20 k_1 \log_{10} |1 + j\tau_1 \omega| - \cdots - 20 k_t \log_{10} |1 + j\tau_t \omega| \end{aligned}$$

Phase

$$\begin{aligned} \angle G(j\omega) = & \angle \tilde{\beta} - k_0 \angle j\omega + \\ & \ell_1 \angle (j\sigma_1 \omega + 1) + \cdots + \ell_r \angle (j\sigma_r \omega + 1) - \\ & k_1 \angle (j\tau_1 \omega + 1) - \cdots - k_t \angle (j\tau_t \omega + 1) \end{aligned}$$

First-order real poles and zeros

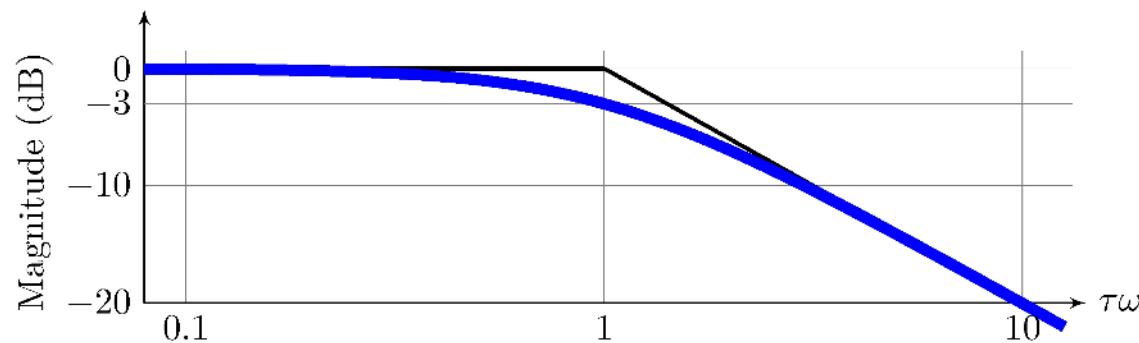
Transfer-function

$$G(s) = \frac{1}{(1 + s\tau)^k}$$

Magnitude

$$20 \log_{10} |G(j\omega)| = -20 k \log_{10} |1 + j\omega\tau| \approx \begin{cases} 0, & |\omega| < |\tau|^{-1} \\ -20 k \log_{10} |\tau| - 20 k \log_{10} |\omega|, & |\omega| \geq |\tau|^{-1} \end{cases}$$

Example: simple pole ($\tau = \pm 1$, $k = 1$)



First-order real poles and zeros

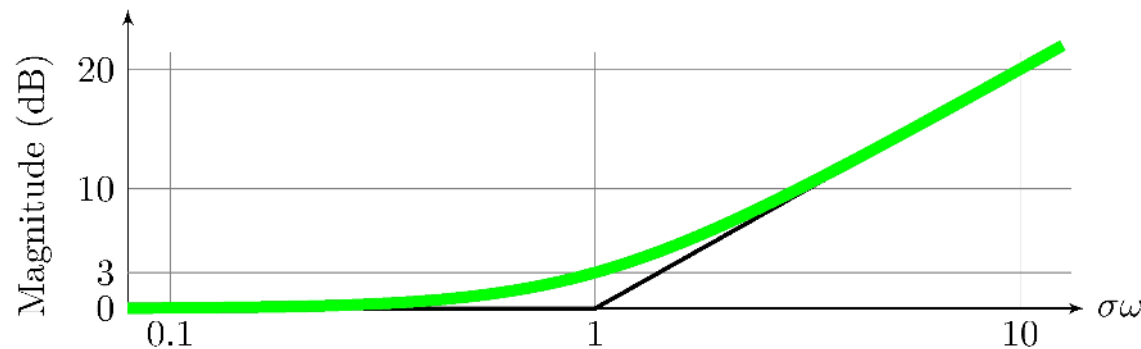
Transfer-function

$$G(s) = (1 + s\tau)^k$$

Magnitude

$$20 \log_{10} |G(j\omega)| = 20 k \log_{10} |1 + j\omega\tau| \approx \begin{cases} 0, & |\omega| < |\tau|^{-1} \\ 20 k \log_{10} |\tau| + 20 k \log_{10} |\omega|, & |\omega| \geq |\tau|^{-1} \end{cases}$$

Example: simple zero ($\tau = \pm 1$, $k = 1$)



First-order real poles and zeros

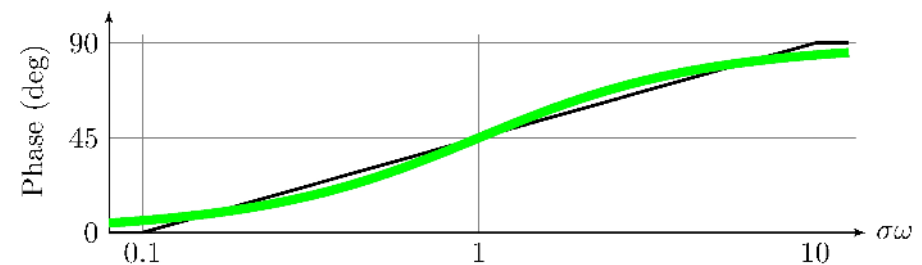
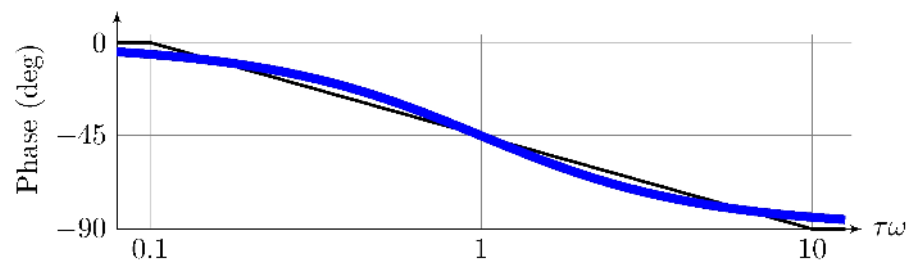
Transfer-function

$$G(s) = \frac{1}{(1 + s\tau)^k}$$

Phase

$$\angle G(j\omega) = -k \angle(1 + j\omega\tau) \approx \frac{\tau}{|\tau|} \begin{cases} 0, & |\omega| < \frac{1}{10} |\tau|^{-1} \\ -\frac{k\pi}{4} \log_{10} 10 |\tau| \omega, & \frac{1}{10} |\tau|^{-1} \leq \omega < 10 |\tau|^{-1} \\ -\frac{k\pi}{2}, & \omega \geq 10 |\tau|^{-1} \end{cases}$$

Example: simple pole ($\tau = 1, k = 1$) Example: simple pole ($\tau = -1, k = 1$)



First-order real poles and zeros

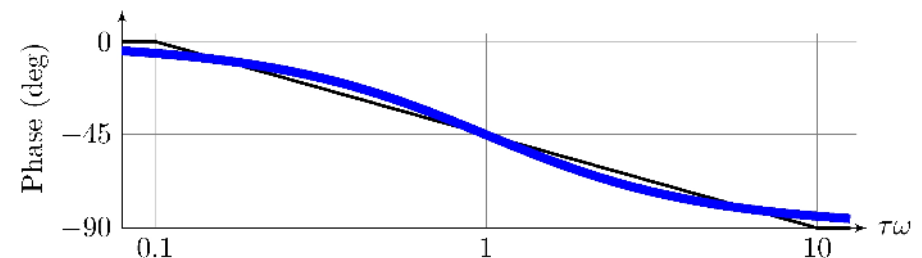
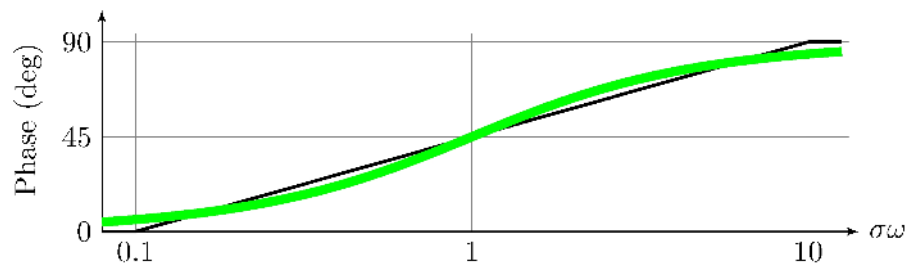
Transfer-function

$$G(s) = (1 + s\tau)^k$$

Phase

$$\angle G(j\omega) = k \angle(1 + j\omega\tau) \approx \frac{\tau}{|\tau|} \begin{cases} 0, & |\omega| < \frac{1}{10} |\tau|^{-1} \\ \frac{k\pi}{4} \log_{10} 10 |\tau| \omega, & \frac{1}{10} |\tau|^{-1} \leq \omega < 10 |\tau|^{-1} \\ \frac{k\pi}{2}, & \omega \geq 10 |\tau|^{-1} \end{cases}$$

Example: simple zero ($\tau = 1, k = 1$) Example: simple zero ($\tau = -1, k = 1$)



Second-order complex poles and zeros

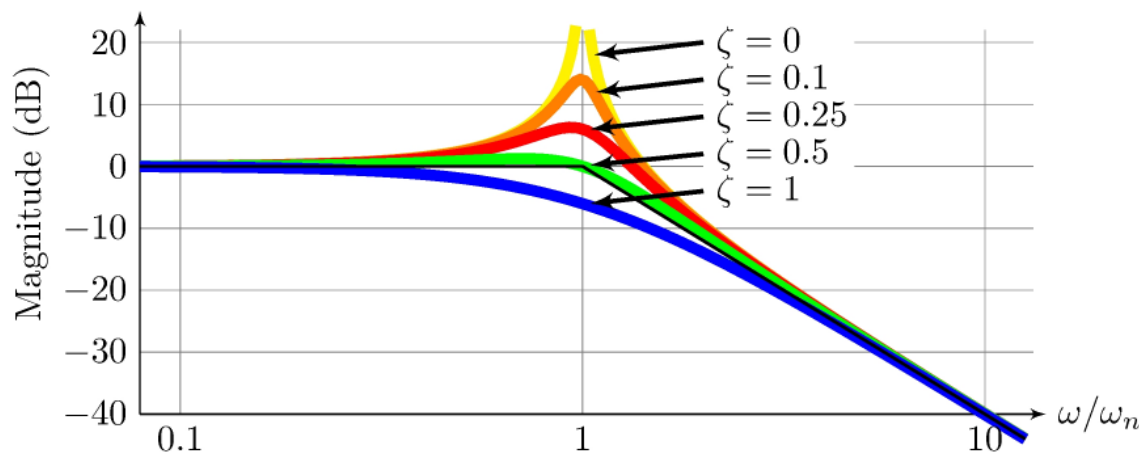
Transfer-function

$$G(s) = \frac{1}{(1 + 2\zeta s/\omega_n + s^2/\omega_n^2)^k}, \quad \omega_n > 0, \quad |\zeta| < 1$$

Magnitude

$$20 \log_{10} |G(j\omega)| = -20k \log_{10} |1 - \omega^2/\omega_n^2 + j2\zeta\omega/\omega_n| \approx \begin{cases} 0, & |\omega| < \omega_n \\ 40k \log_{10} |\omega_n| - 40k \log_{10} |\omega|, & |\omega| \geq \omega_n \end{cases}$$

Example: complex pair of poles ($\omega_n = 1, k = 1$)



Second-order complex poles and zeros

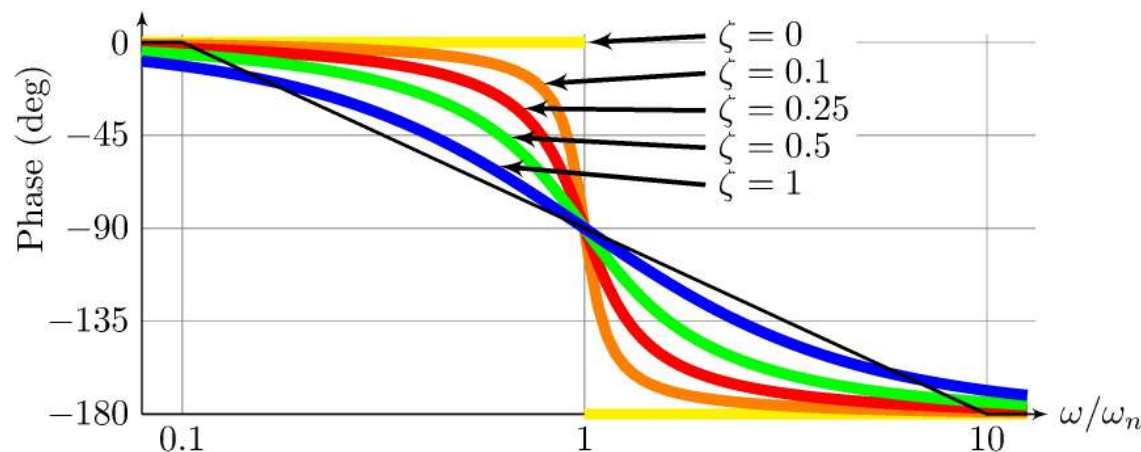
Transfer-function

$$G(s) = \frac{1}{(1 + 2\zeta s/\omega_n + s^2/\omega_n^2)^k}, \quad \omega_n > 0, \quad |\zeta| < 1$$

Phase

$$\angle G(j\omega) = k \angle 1 - \omega^2/\omega_n^2 + j2\zeta\omega/\omega_n \approx \frac{\zeta}{|\zeta|} \begin{cases} 0, & |\omega| < \frac{1}{10} \omega_n \\ -\frac{k\pi}{2} \log_{10} 10\omega/\omega_n & \frac{1}{10} \omega_n \leq \omega < 10\omega_n \\ -k\pi, & \omega \geq 10\omega_n \end{cases}$$

Example: complex pair of poles ($\omega_n = 1, k = 1$)



Poles and zeros at the origin

Transfer-function

$$G(s) = \frac{1}{s^k}$$

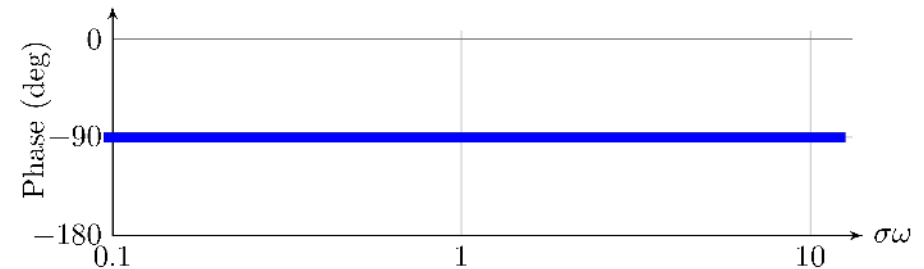
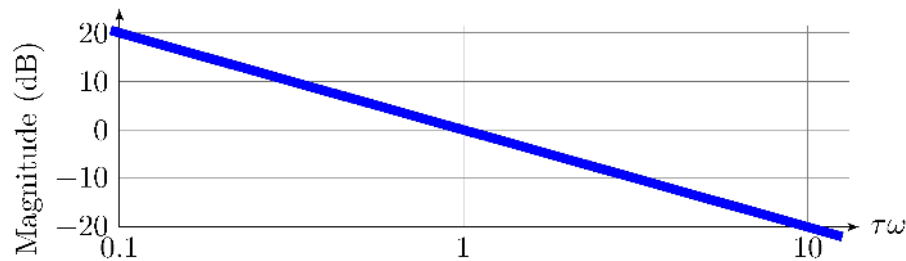
Magnitude

$$20 \log_{10} |G(j\omega)| = -20 k \log_{10} |\omega|$$

Phase

$$\angle G(j\omega) = -k \frac{\pi}{2}$$

Example: simple pole ($k = 1$)



Example: C_{lead} from Section 6.5

Transfer-function

$$G(s) = C_{\text{lead}}(s) = 3.83 \frac{s + 7.5}{s + 21}$$

Normalized

$$G(s) = \tilde{\beta} \frac{1 + s/7.5}{1 + s/21}, \quad \tilde{\beta} = \frac{3.83 \times 7.5}{21} \approx 1.37$$

Magnitude slopes

$$\begin{array}{ll} 0\text{dB/decade}, & 0 \leq \omega \leq 7.5 \\ +20\text{dB/decade}, & 7.5 \leq \omega \leq 21 \\ 0\text{dB/decade}, & \omega \geq 21 \end{array}$$

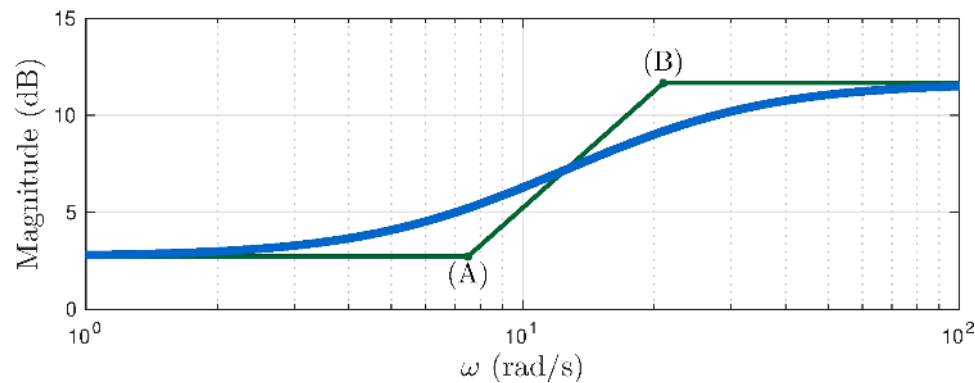
Magnitude key points

$$\begin{array}{ll} 20 \log_{10} 1.37 \approx 2.7\text{dB}, & \omega \leq 7.5 \quad (\text{A}) \\ 2.7\text{dB} + 20 \log_{10} 21/7.5 \approx 11.6\text{dB}, & \omega \geq 21 \quad (\text{B}) \end{array}$$

Example: C_{lead} from Section 6.5

Transfer-function

$$G(s) = C_{\text{lead}}(s) = 3.83 \frac{s + 7.5}{s + 21}$$



Magnitude slopes

0dB/decade, $0 \leq \omega \leq 7.5$
 +20dB/decade, $7.5 \leq \omega \leq 21$
 0dB/decade, $\omega \geq 21$

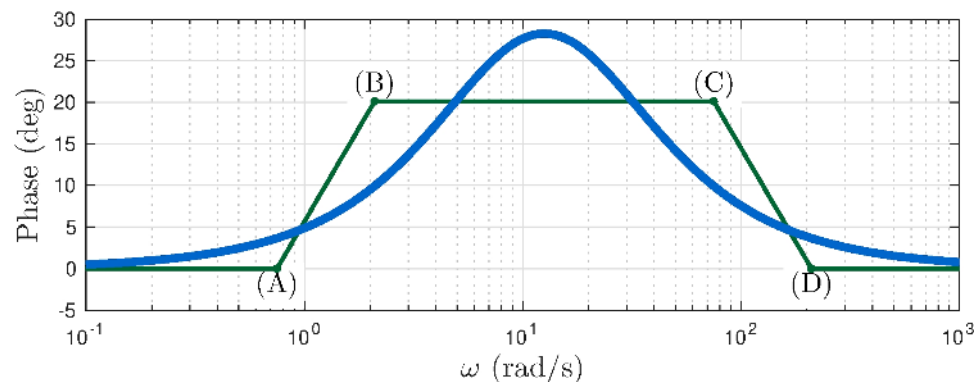
Magnitude key points

$20 \log_{10} 1.37 \approx 2.7\text{dB}, \quad \omega \leq 7.5 \quad (\text{A})$
 $2.7\text{dB} + 20 \log_{10} 21/7.5 \approx 11.6\text{dB}, \quad \omega \geq 21 \quad (\text{B})$

Example: C_{lead} from Section 6.5

Transfer-function

$$G(s) = C_{\text{lead}}(s) = 1.37 \frac{1 + s/7.5}{1 + s/21}$$



Phase slopes

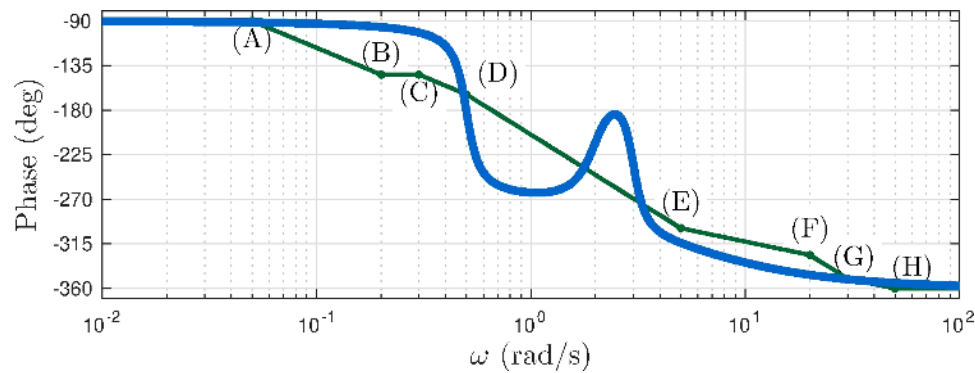
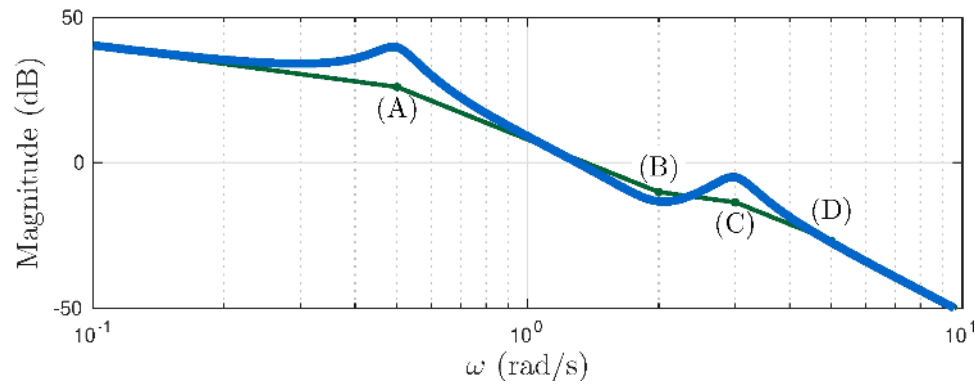
$0^\circ/\text{decade},$	$0 \leq \omega \leq 0.75$
$+45^\circ/\text{decade},$	$0.75 \leq \omega \leq 2.1$
$0^\circ/\text{decade},$	$2.1 \leq \omega \leq 75$
$-45^\circ/\text{decade},$	$75 \leq \omega \leq 210$
$0^\circ/\text{decade},$	$\omega \geq 210$

Phase key points

$\approx 0^\circ,$	$\omega = 0.75$	(A)
$45^\circ \log_{10} 2.1/0.75 \approx 20^\circ,$	$\omega = 2.1,$	(B)
$45^\circ \log_{10} 2.1/0.75 \approx 20^\circ,$	$\omega = 75,$	(C)
$0^\circ \log_{10} 2.1/0.75 \approx 0^\circ,$	$\omega = 210.$	(C)

See book for a more challenging example

$$G(s) = \frac{28.1s^2 + 22.4s + 112.4}{s^6 + 5.7s^5 + 12.81s^4 + 47.6s^3 + 7.5s^2 + 11.25s}$$



7.2 Non-Minimum-Phase Systems

A simple example

Frequency responses of

$$G_1 = 1, \quad G_2 = -1, \quad G_3 = \frac{s - 1}{s + 1}$$

have the exact same magnitude

$$|G_1(j\omega)| = |G_2(j\omega)| = |G_3(j\omega)| = 1$$

but different phases

$$\angle G_1(j\omega) = 0, \quad \angle G_2(j\omega) = \pi, \quad \angle G_3(j\omega) = \pi - 2 \tan^{-1} \omega$$

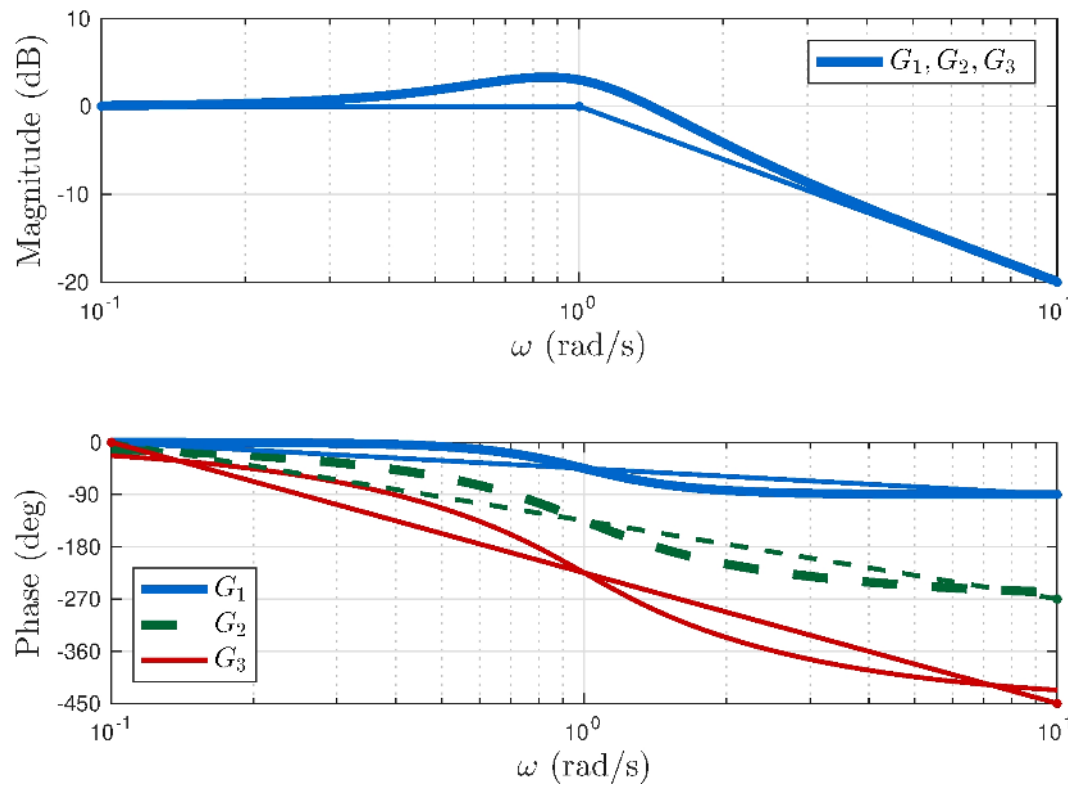
G_1 is **minimum phase**

A more complex example

Frequency response of

$$G_1 = \frac{1+s}{s^2+s+1}, \quad G_2 = \frac{1-s}{s^2+s+1}, \quad G_3 = \frac{(1-s)^2}{s^3+2s^2+2s+1}$$

Bode plots



A more complex example

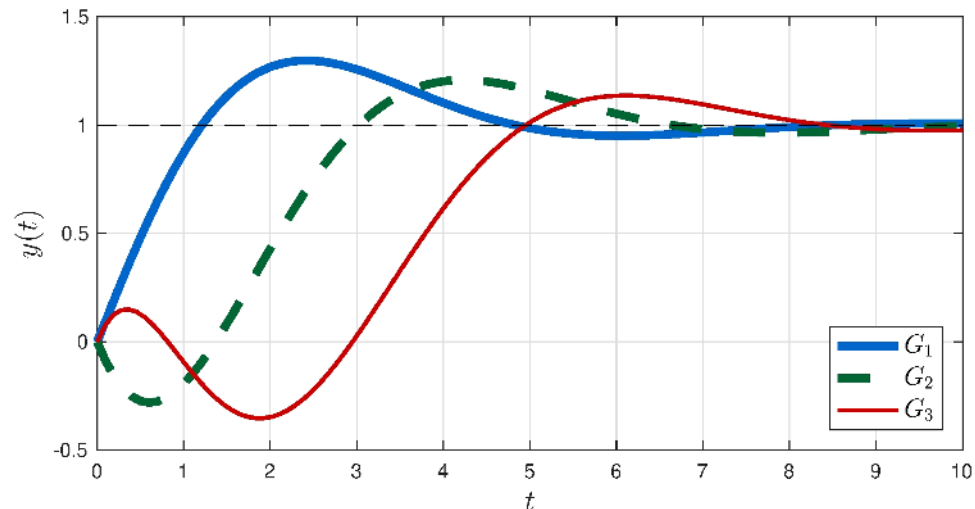
Frequency response of

$$G_1 = \frac{1+s}{s^2+s+1}, \quad G_2 = \frac{1-s}{s^2+s+1}, \quad G_3 = \frac{(1-s)^2}{s^3+2s^2+2s+1}$$

have the exact same magnitude $|G_1(j\omega)| = |G_2(j\omega)| = |G_3(j\omega)|$

G_1 is minimum phase

Step response



Other examples

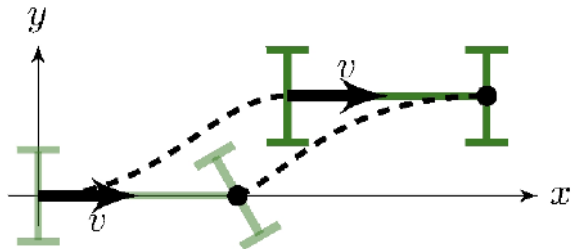
Systems with delays

Padé approximation of order n

$$e^{-s\tau} \approx \frac{1 - a_1 s + a_2 s^2 - \cdots \pm a_n s^n}{1 + a_1 s + a_2 s^2 + \cdots + a_n s^n}$$

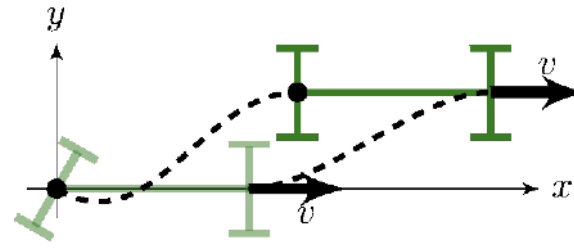
Steering vehicles

Forward lane-change



Minimum-phase

Backward lane-change



Non-minimum-phase

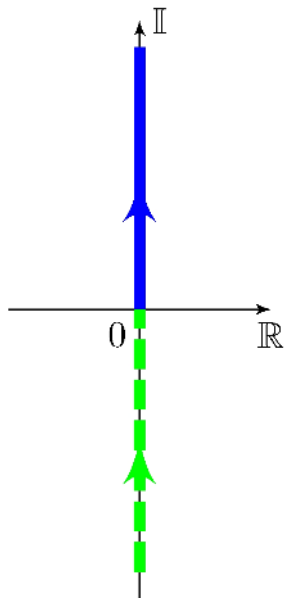
7.3 Polar Plots

Parametric curve

$$s = G(j\omega), \quad \omega \in \mathbb{R}$$

Image of the imaginary axis under the mapping G

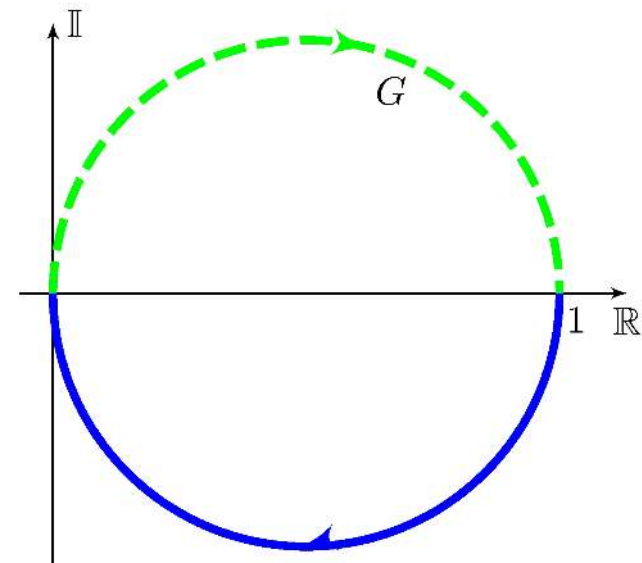
Imaginary axis



Mapping

$$G(s) = \frac{1}{s+1}$$

Image



From Bode plots to polar plots

Example

$$G(s) = (s + 1)^{-1}$$

Polar plot

$$G(j\omega) = \frac{1}{j\omega + 1} = \frac{1}{2} + \frac{1}{2} \frac{1 - j\omega}{1 + j\omega} = \frac{1}{2} + \frac{1}{2} e^{j\theta}, \quad -\pi < \theta = -2 \tan^{-1} \omega < \pi$$

Bode plots

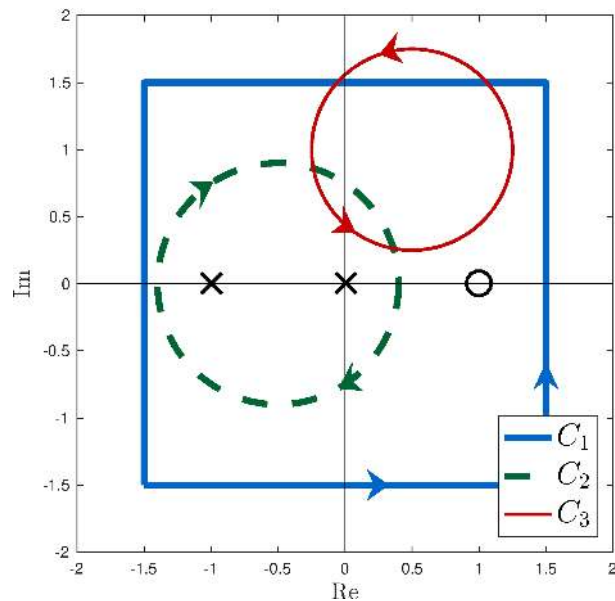
Polar plot

7.4 The Argument Principle

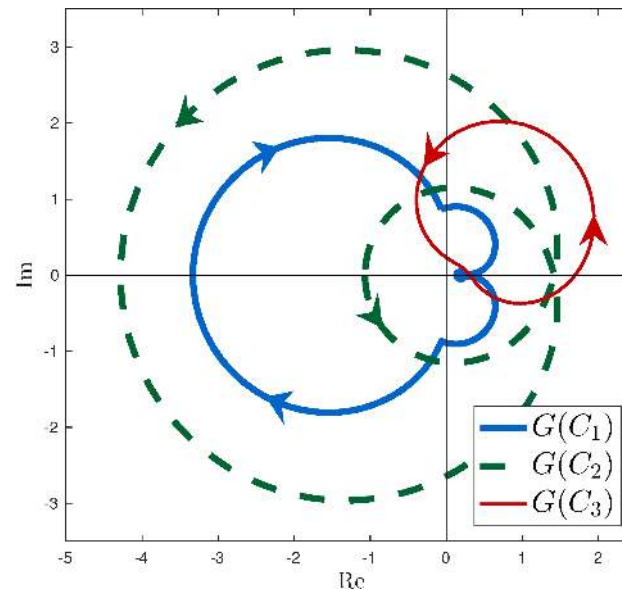
Total argument variation

$\Delta_C^0 \arg G(s)$: total argument variation of the image of C under G about the origin

Contours



Images



Total argument

$$\frac{1}{2\pi} \Delta_{C_1}^0 \arg G(s) = -1$$

$$\frac{1}{2\pi} \Delta_{C_2}^0 \arg G(s) = 2$$

$$\frac{1}{2\pi} \Delta_{C_3}^0 \arg G(s) = 0$$

The Argument Principle

Theorem 7.1

$$\frac{1}{2\pi} \Delta_C^0 \arg f(s) = \frac{1}{2\pi j} \int_C \frac{f'(s)}{f(s)} ds = Z_C - P_C$$

Z_C is the number of zeros inside C

P_C is the number of poles inside C

Assumptions

C is a positively oriented simple closed contour

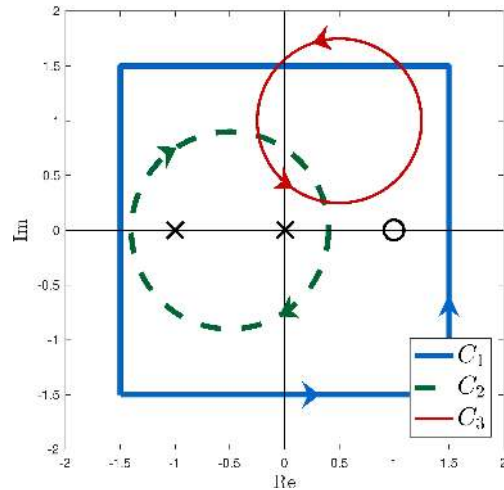
f has no zeros on C

f is analytic inside and on C except at a finite number of poles inside C

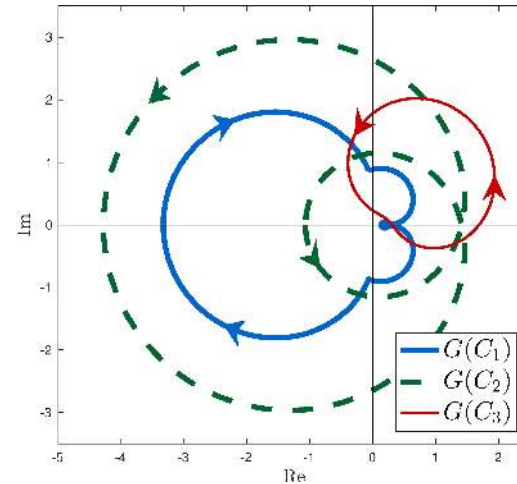
If contour is negatively oriented

$$\frac{1}{2\pi} \Delta_C^0 \arg f(s) = -\frac{1}{2\pi} \Delta_{-C}^0 \arg f(s) = P_C - Z_C$$

Contours



Images



Poles and zeros

$$\begin{aligned} Z_{C_1} &= 1, & P_{C_1} &= 2 \\ Z_{C_2} &= 0, & P_{C_2} &= 2 \\ Z_{C_3} &= 0, & P_{C_3} &= 0 \end{aligned}$$

Total argument

$$\begin{aligned} (1/2\pi)\Delta_{C_1}^0 \arg G(s) &= Z_{C_1} - P_{C_1} = -1 \\ (1/2\pi)\Delta_{C_2}^0 \arg G(s) &= P_{C_2} - Z_{C_2} = 2 \\ (1/2\pi)\Delta_{C_3}^0 \arg G(s) &= Z_{C_3} - P_{C_3} = 0 \end{aligned}$$

Application

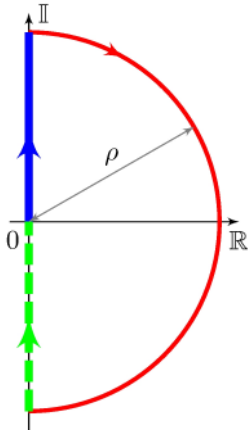
Testing for asymptotic stability

7.5 Stability in the Frequency Domain

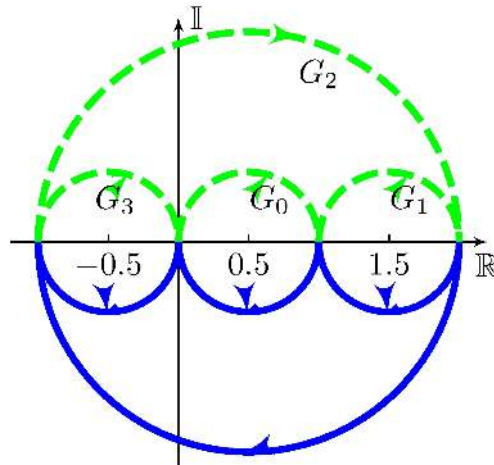
Basic idea

$$P_{\Gamma} = Z_{\Gamma} + \frac{1}{2\pi} \Delta_{\Gamma}^0 \arg G(s)$$

Contour Γ



Polar plot



Transfer-functions

$$G_0(s) = \frac{1}{s+1}$$

$$G_1(s) = \frac{s+2}{s+1} = 1 + G_0(s)$$

$$G_2(s) = \frac{2-s}{s+1} = 3G_0(s) - 1$$

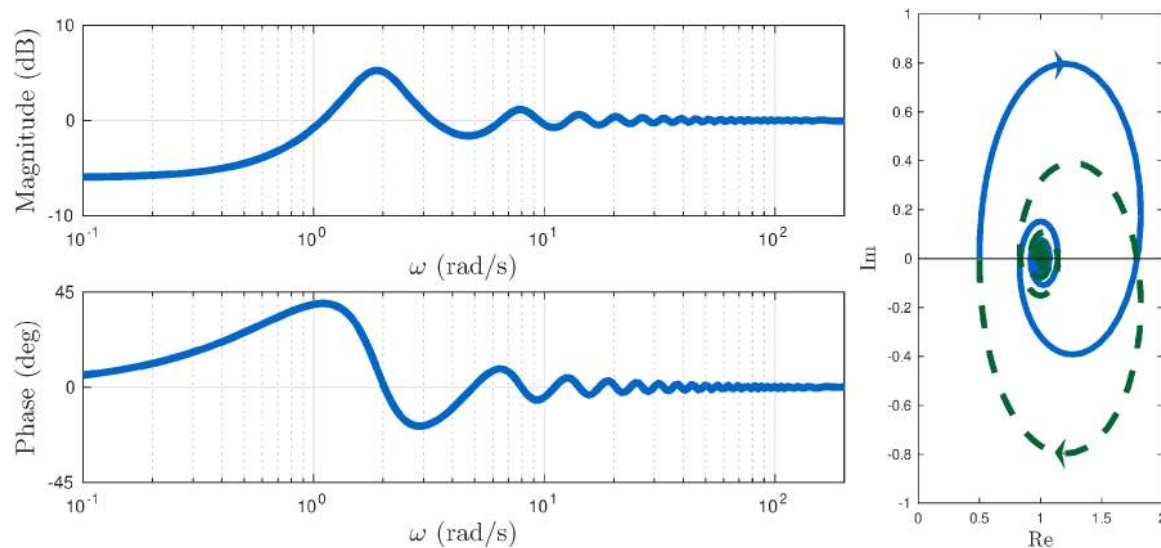
$$G_3(s) = \frac{-s}{s+1} = G_0(s) - 1$$

$$\begin{aligned} Z_{\Gamma i} &= 0, & (2\pi)^{-1} \Delta_{\Gamma}^0 \arg G_i(s) &= 0, & P_{\Gamma i} &= 0 + 0 = 0, & i &= \{0, 1, 3\} \\ Z_{\Gamma 2} &= 1, & (2\pi)^{-1} \Delta_{\Gamma}^0 \arg G_2(s) &= -1, & P_{\Gamma 2} &= 1 - 1 = 0 \end{aligned}$$

Features

Attention near the origin if G is strictly proper

Can be used even if G is not rational, e.g. $G(s) = (s + 1)/(s + 1 + e^{-s})$

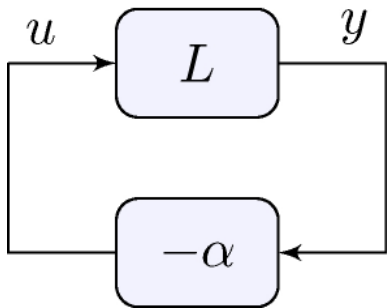


More impressive application

Closed-loop stability analysis: the Nyquist stability criterion

7.6 Nyquist Stability Criterion

Basic idea



Closed loop poles are the zeros of the characteristic equation

$$1 + \alpha L(s) = 0, \quad \alpha > 0$$

which are the same as the zeros of

$$L_\alpha(s) = \frac{1}{\alpha} + L(s), \quad \alpha > 0$$

Image of Γ under $L_\alpha(s)$ is the same as image of Γ under $L(s)$ shifted by $\frac{1}{\alpha}$

Example #1

Transfer-function

$$L(s) = \frac{s - 0.5}{s^4 + 2.5s^3 + 3s^2 + 2.5s + 1}, \quad L_\alpha(s) = \frac{1}{\alpha} + L(s)$$

Image of Γ under L

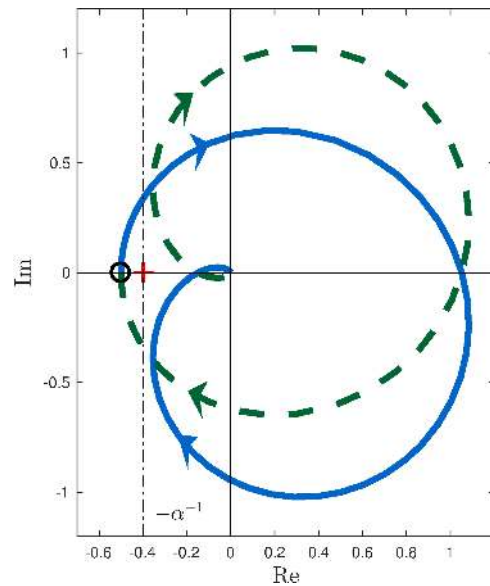
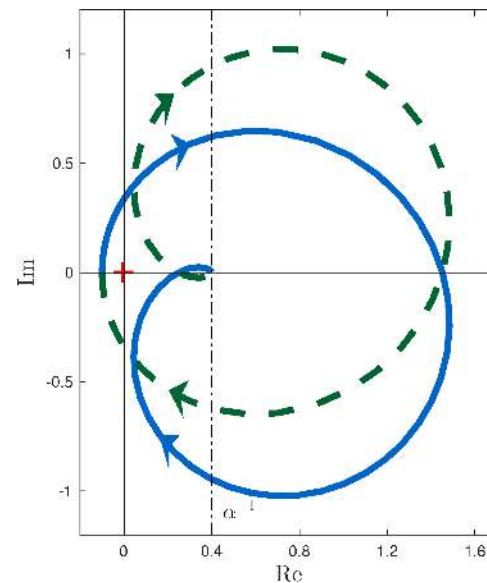


Image of Γ under L_α , $\alpha = 2.5$



Basic idea

Image of Γ under $L_\alpha(s)$ is the same as image of Γ under $L(s)$ shifted by $\frac{1}{\alpha}$

Total argument variation are the same if we shift the origin

$$\frac{1}{2\pi} \Delta_\Gamma^0 \arg L_\alpha(s) = \frac{1}{2\pi} \Delta_\Gamma^{-1/\alpha} \arg L(s) = P_\Gamma - Z_\Gamma$$

Theorem 7.2 (Nyquist)

Closed-loop asymptotic stability if and only if

$$Z_\Gamma = P_\Gamma - \frac{1}{2\pi} \Delta_\Gamma^{-1/\alpha} \arg L(s) = 0$$

P_Γ : # of poles of $L(s)$ in Γ (unstable open-loop poles)

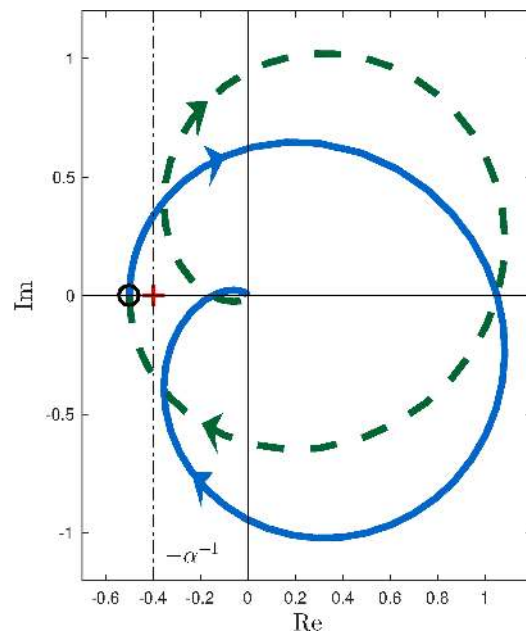
Z_Γ : # of poles of $1 + \alpha L(s)$ in Γ (unstable closed-loop poles)

Example #1

Transfer-function

$$L(s) = \frac{s - 0.5}{s^4 + 2.5s^3 + 3s^2 + 2.5s + 1}$$

Image of Γ under L



Encirclements of $-1/\alpha = -0.4$

$$\frac{1}{2\pi} \Delta_{\Gamma}^{-0.4} \arg L_{\alpha}(s) = -1$$

Closed-loop asymptotic stability? NO

$$Z_{\Gamma} = P_{\Gamma} - \frac{1}{2\pi} \Delta_{\Gamma}^0 \arg L_{\alpha}(s) = 0 - (-1) = 1$$

closed-loop poles in Γ

What if $-\alpha^{-1} < -0.5$?

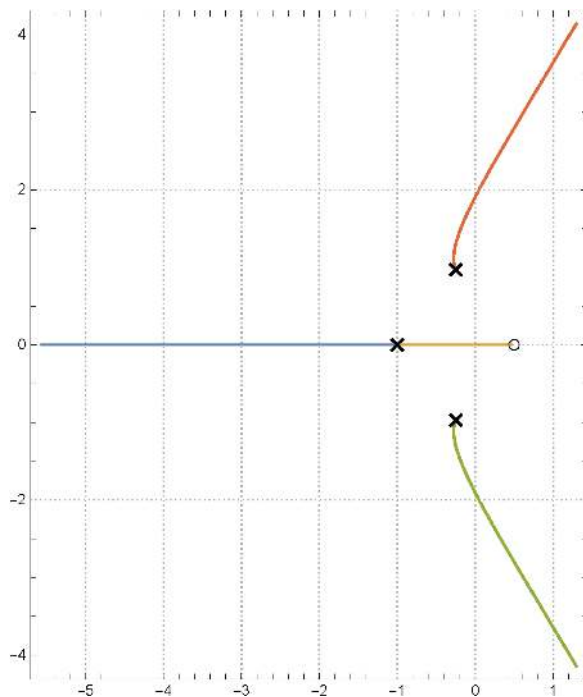
Or if $-0.1 < -\alpha^{-1} < 0$?

Example #1

Transfer-function

$$L(s) = \frac{s - 0.5}{s^4 + 2.5s^3 + 3s^2 + 2.5s + 1}$$

Root-locus

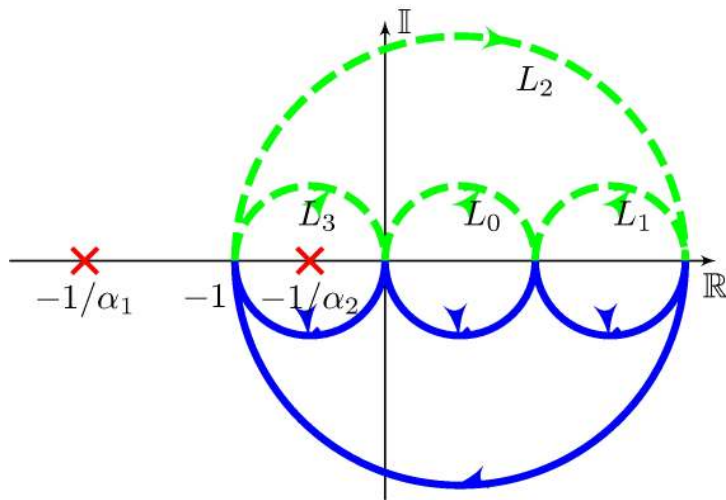


Example #2

Transfer-functions

$$L_0(s) = \frac{1}{s+1}, \quad L_1(s) = \frac{s+2}{s+1}, \quad L_2(s) = \frac{2-s}{s+1}, \quad L_3(s) = \frac{-s}{s+1}$$

Nyquist plot



Closed-loop stability

$$P_\Gamma = 0$$

$$L_0 \text{ and } L_1$$

$$\frac{1}{2\pi} \Delta_\Gamma^{-1/\alpha} \arg L_\alpha(s) = 0 \quad \implies \quad Z_\Gamma = 0$$

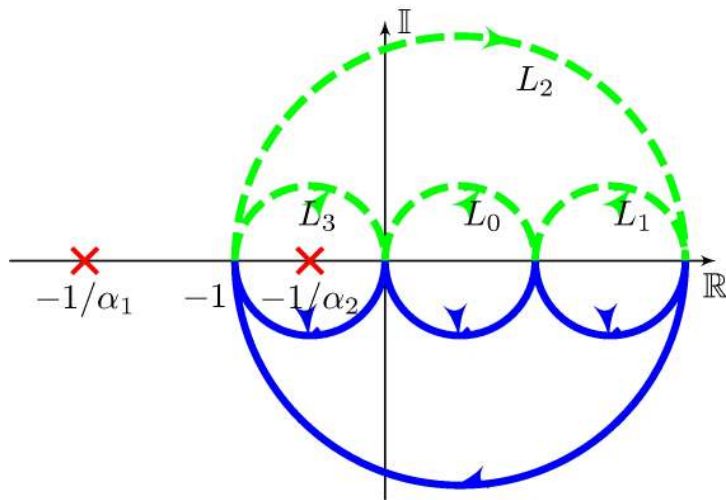
stable for all $\alpha > 0$

Example #2

Transfer-functions

$$L_0(s) = \frac{1}{s+1}, \quad L_1(s) = \frac{s+2}{s+1}, \quad L_2(s) = \frac{2-s}{s+1}, \quad L_3(s) = \frac{-s}{s+1}$$

Nyquist plot



Closed-loop stability

$$P_\Gamma = 0$$

L_0 and L_1 stable for all $\alpha > 0$

L_2 and L_3

$$\frac{1}{2\pi} \Delta_\Gamma^{-1/\alpha_1} \arg L_\alpha(s) = 0 \quad \implies \quad Z_\Gamma = 0$$

$$\frac{1}{2\pi} \Delta_\Gamma^{-1/\alpha_2} \arg L_\alpha(s) = -1 \quad \implies \quad Z_\Gamma = 1$$

stable for all $\alpha = \alpha_1$, unstable for all $\alpha = \alpha_2$

Nyquist Stability Criterion

Theorem 7.2 (Nyquist)

Closed-loop asymptotic stability if and only if

$$Z_{\Gamma} = P_{\Gamma} - \frac{1}{2\pi} \Delta_{\Gamma}^{-1/\alpha} \arg L(s) = 0$$

P_{Γ} : # of poles of G in Γ (unstable open-loop poles)

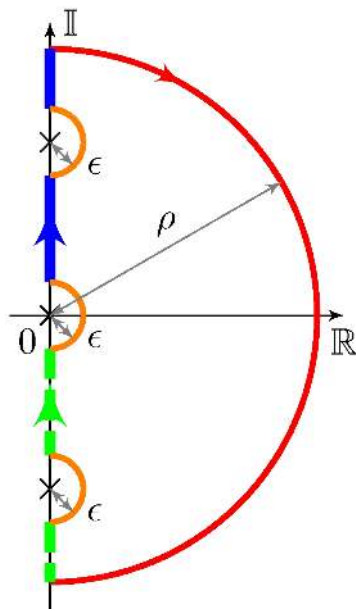
Z_{Γ} : # of poles of $1 + \alpha L(s)$ in Γ (unstable closed-loop poles)

Poles on the imaginary axis can be dealt with indentations

Nyquist Stability Criterion

Indented Γ contour

$$L(s) = \frac{F(s)}{(s - j\omega_0)^n}, \quad F(j\omega_0) \neq 0$$



Magnitude

$$|L(j\omega_0 + \epsilon e^{j\theta})| \approx \epsilon^{-n} |F(j\omega_0)| \rightarrow \infty, \quad \epsilon \approx 0$$

Phase

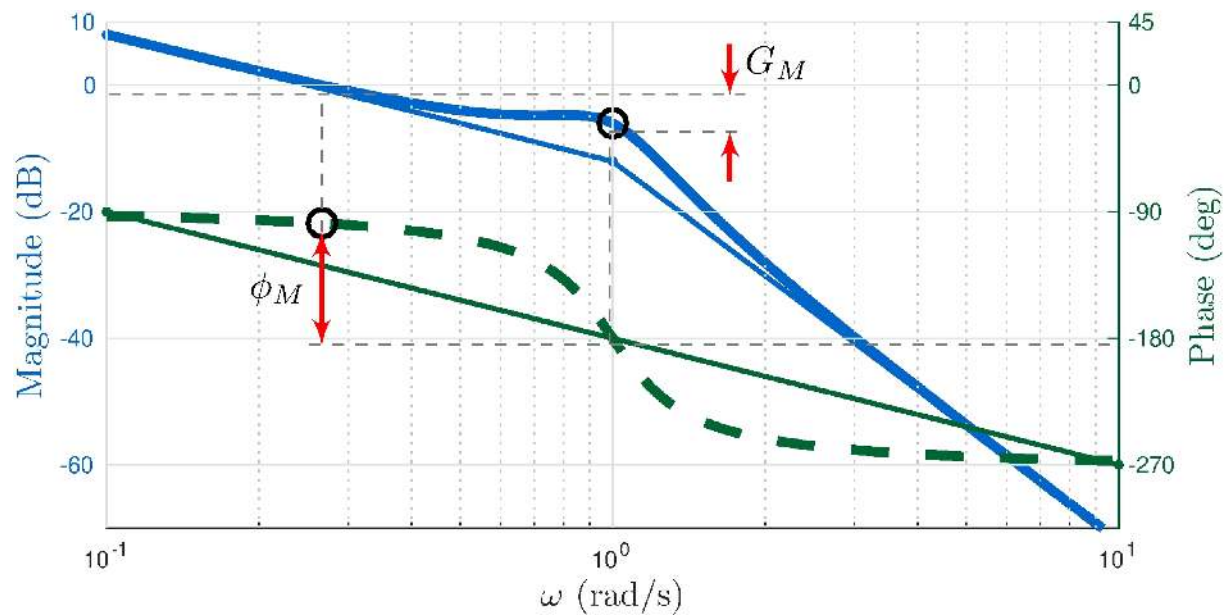
$$\angle L(j\omega_0 + \epsilon e^{j\theta}) \approx \angle F(j\omega_0) - n\theta, \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, \quad \epsilon \approx 0$$

Example #3

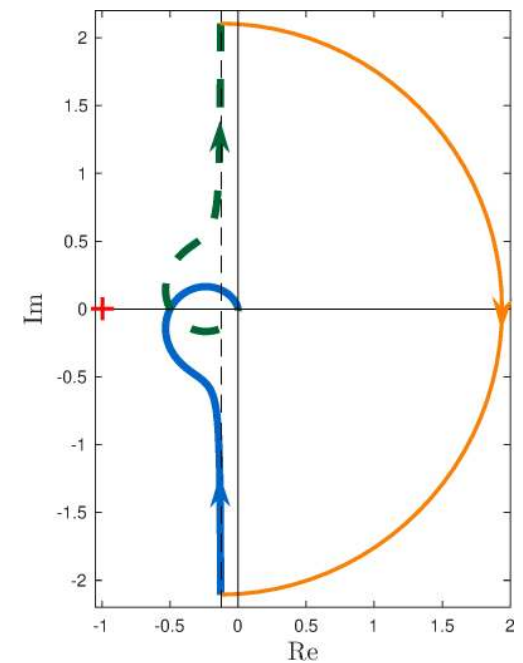
Transfer-function

$$L(s) = \frac{1}{4} \frac{1}{s(s^2 + s/2 + 1)}, \quad \text{stable for } \alpha = 1$$

Bode plots



Nyquist plot

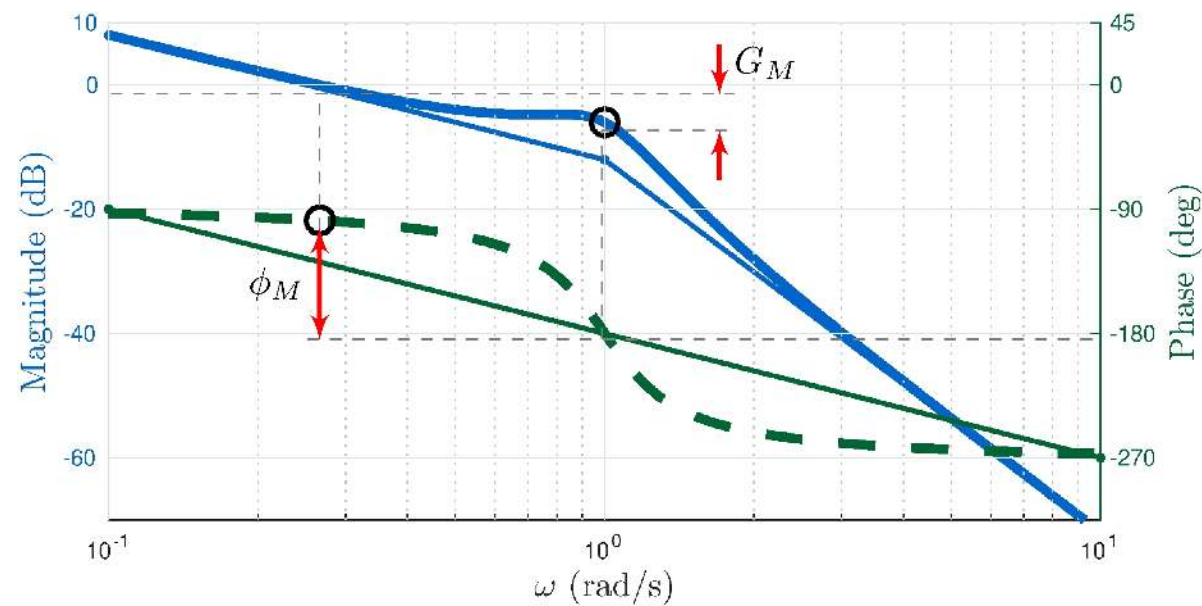


7.7 Stability Margins

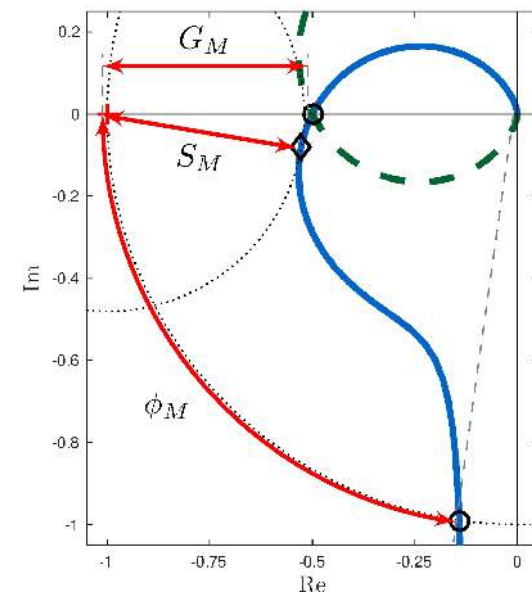
Gain margin

$$G_M = |L(j\omega_G)|^{-1}, \quad \text{for } \omega_G \text{ s.t. } \angle L(j\omega_G) = \pi$$

Bode plots



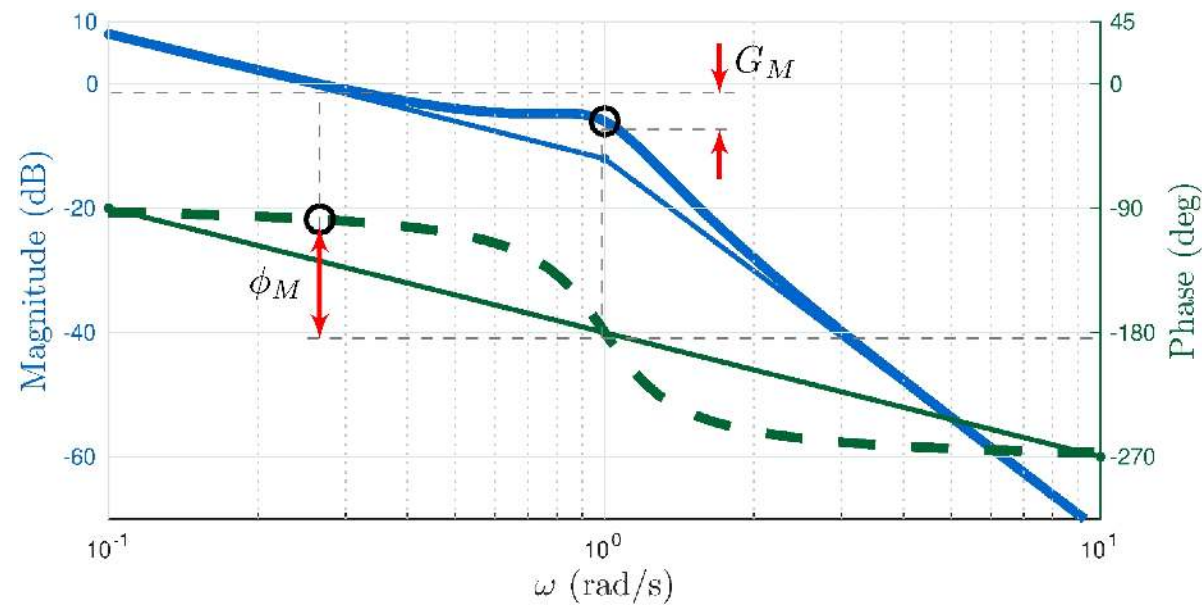
Nyquist plot



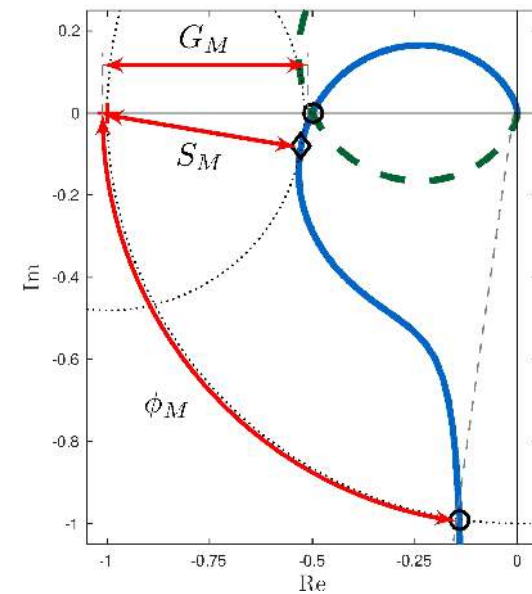
Phase margin

$$\phi_M = \angle L(j\omega_\phi) - \pi, \quad \text{for } \omega_\phi \text{ s.t. } |L(j\omega_\phi)| = 1$$

Bode plots



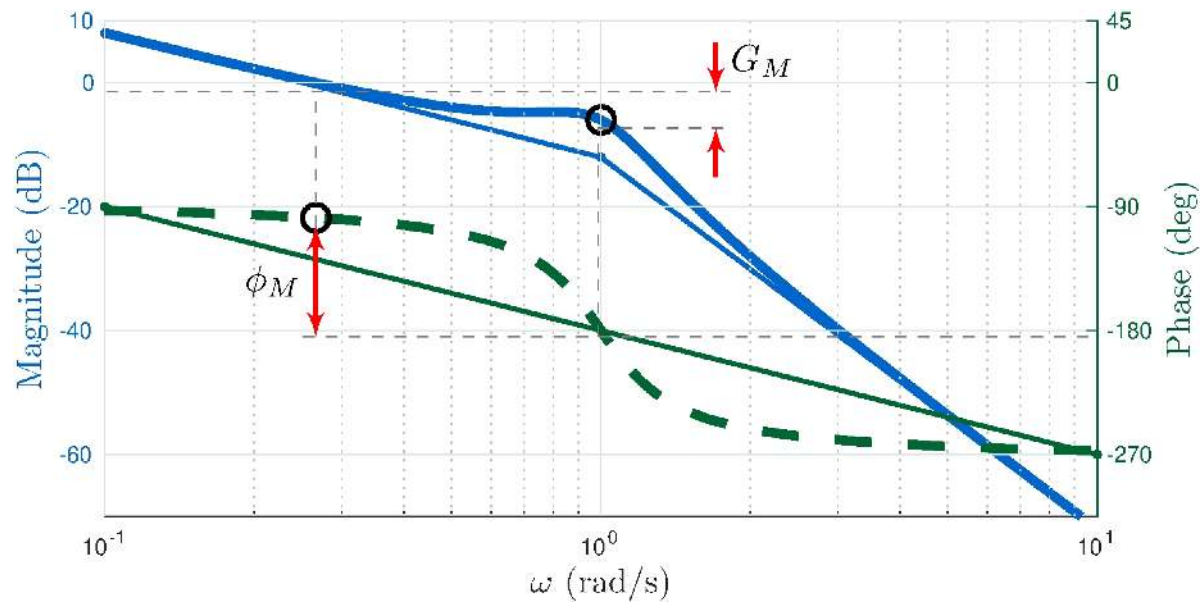
Nyquist plot



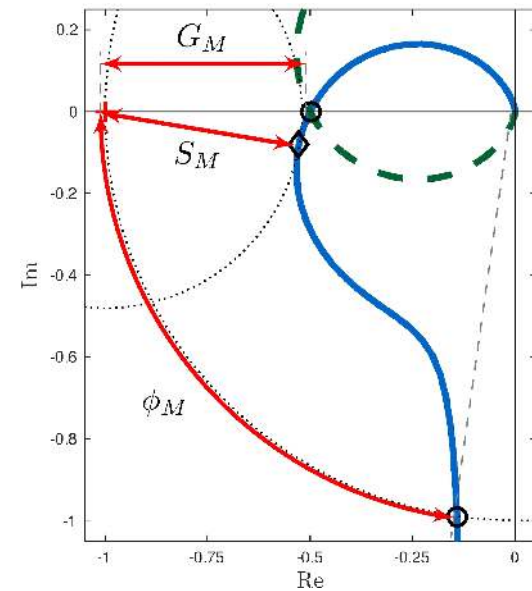
Stability margin

$$S_M = \inf_{\omega} |L(j\omega) - (-1)| = \frac{1}{\sup_{\omega} |S(j\omega)|} = \|S\|_{\infty}^{-1}$$

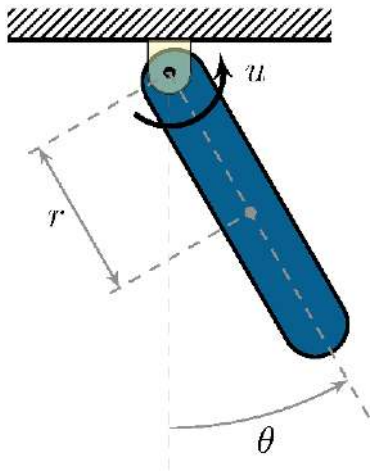
Bode plots



Nyquist plot



7.8 Control of the Simple Pendulum - Part II



Data

$$\begin{aligned}
 m &= 0.5\text{kg}, & \ell &= 0.3\text{m}, & r &= \ell/2 = 0.15\text{m}, \\
 b &= 0\text{kg/s}, & g &= 9.8\text{m/s}^2, & J &= 3.75 \times 10^{-3}\text{kg m}^2, & J_r &= J + mr^2.
 \end{aligned}$$

Linearized models

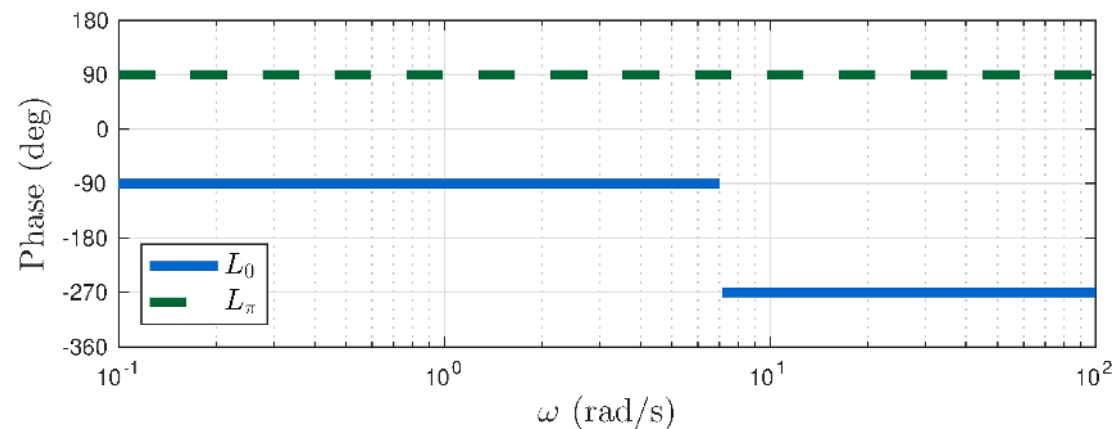
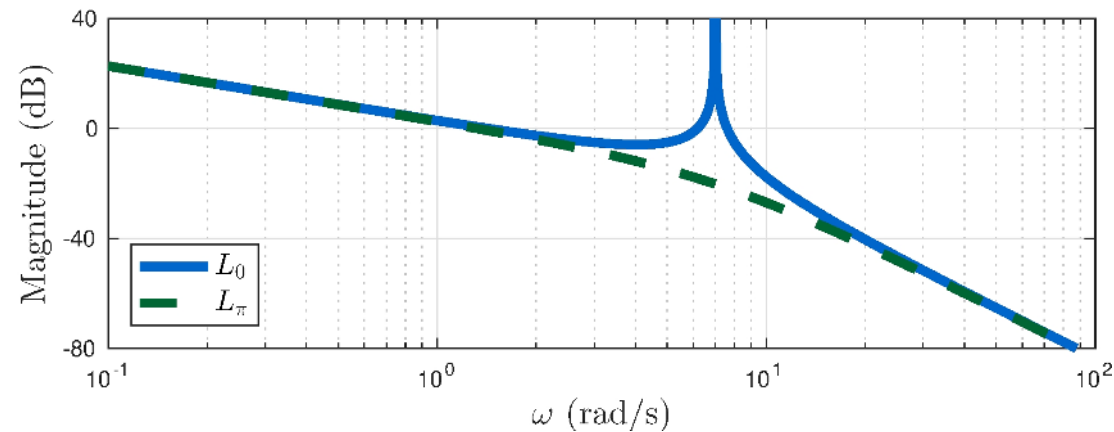
$$G_0(s) = \frac{1/J_r}{s^2 + (b/J_r)s + mgr/J_r} \approx \frac{66.7}{s^2 + 49},$$

$$G_\pi(s) = \frac{1/J_r}{s^2 + (b/J_r)s - mgr/J_r} \approx \frac{66.7}{s^2 - 49}$$

Goal

Design a controller with integral action

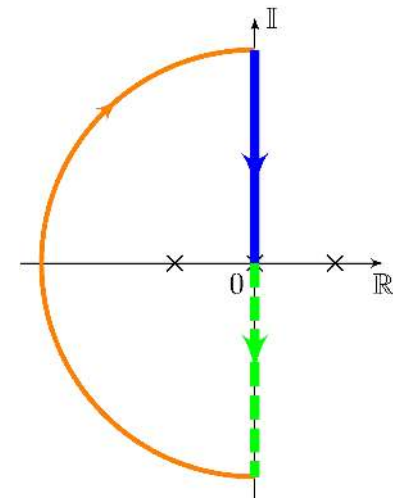
Control of the Simple Pendulum - Part II



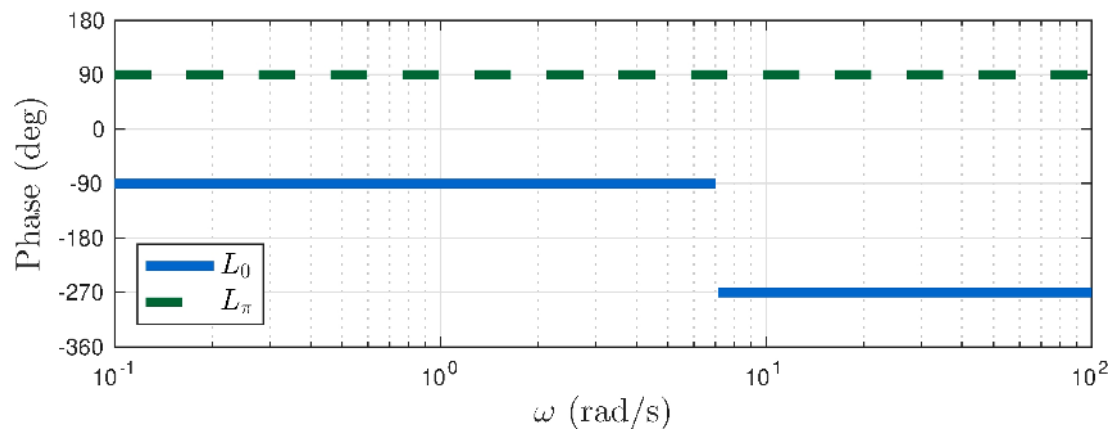
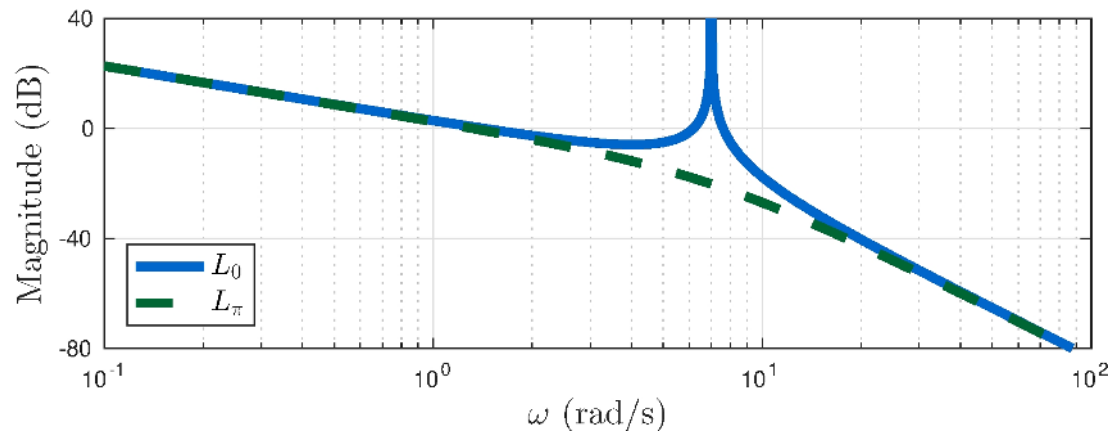
Nyquist analysis

$$L_\pi = \frac{G_\pi}{s} \approx \frac{-1.36}{s(1 + s/7)(1 - s/7)}$$

$$\begin{aligned} Z_\Gamma &= P_\Gamma - \frac{1}{2\pi} \Delta_\Gamma^{-1} \arg L_\pi(s) \\ &= 1 - (-1) = 2 \end{aligned}$$



Control of the Simple Pendulum - Part II



Stability analysis

Because $P_\Gamma = 1$, $L_\pi(j\omega)$ needs

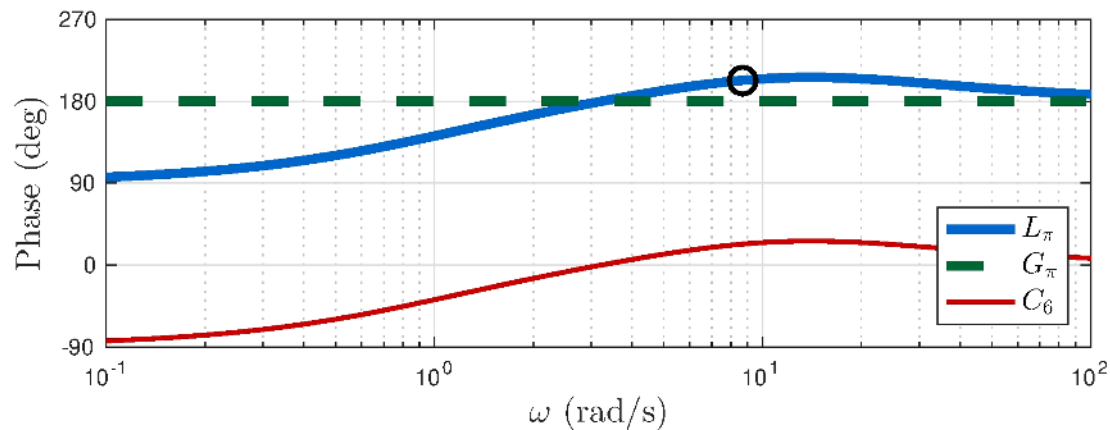
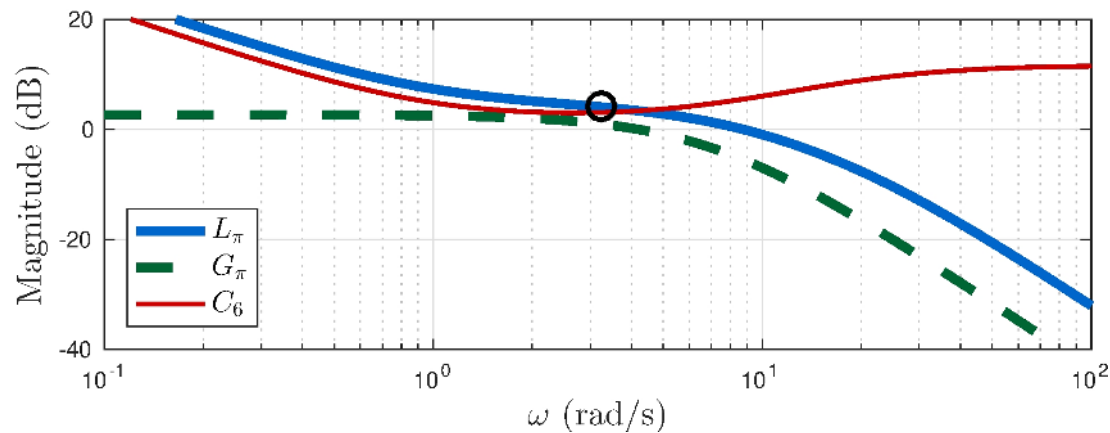
- one counter-clockwise encirclement
- to cross negative real axis
- magnitude greater than one

Counter-clockwise encirclement requires more than 90° phase to be added to $L_\pi(j\omega)$.

Controller must have

- at least two stable zeros
- at least one pole for realizability (in addition to the integrator)

Control of the Simple Pendulum - Part II



Controller candidate

$$C(s) = K \frac{(s + z_1)(s + z_2)}{s(s + p_1)}$$

with

$$p_1 > z_2 \geq z_1$$

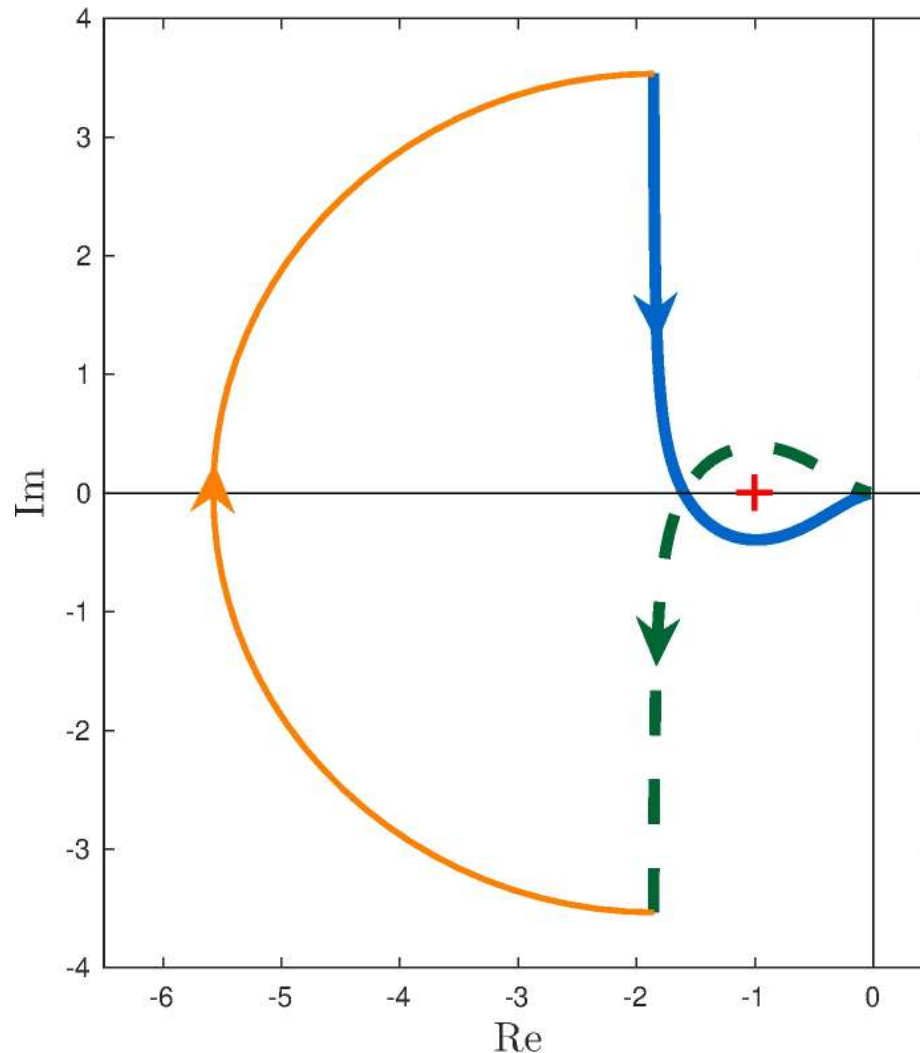
From C_6

$$\begin{aligned} p_1 &= 21, & K &= 3.83 \\ z_2 &= 6.86, & z_1 &= 0.96 \end{aligned}$$

Margins

$$\begin{aligned} G_M &\approx 0.63 \approx -4\text{dB}, & \phi_M &\approx 23^\circ \\ S_M &= 0.40 \end{aligned}$$

Control of the Simple Pendulum - Part II



Controller candidate

$$C(s) = K \frac{(s + z_1)(s + z_2)}{s(s + p_1)}$$

with

$$p_1 > z_2 \geq z_1$$

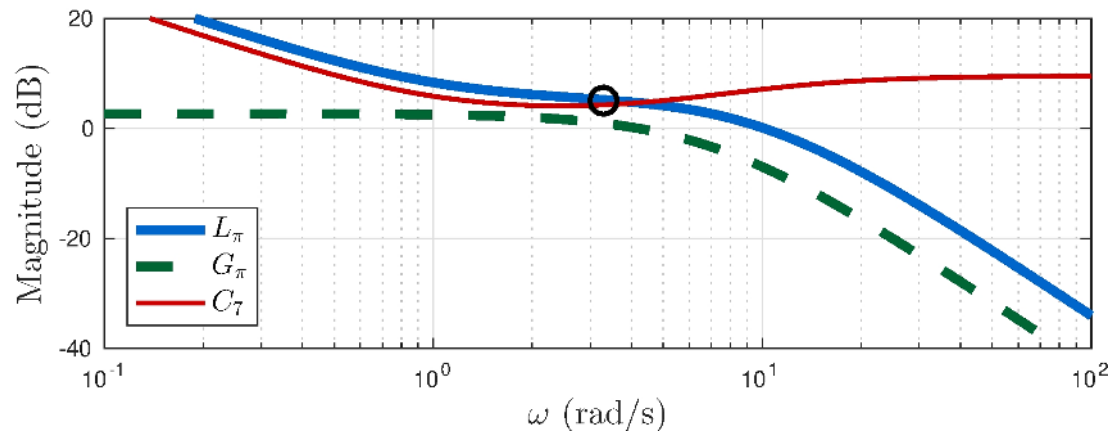
From C_6

$$\begin{aligned} p_1 &= 21, & K &= 3.83 \\ z_2 &= 6.86, & z_1 &= 0.96 \end{aligned}$$

Margins

$$\begin{aligned} G_M &\approx 0.63 \approx -4\text{dB}, & \phi_M &\approx 23^\circ \\ S_M &= 0.40 \end{aligned}$$

Control of the Simple Pendulum - Part II



Controller candidate

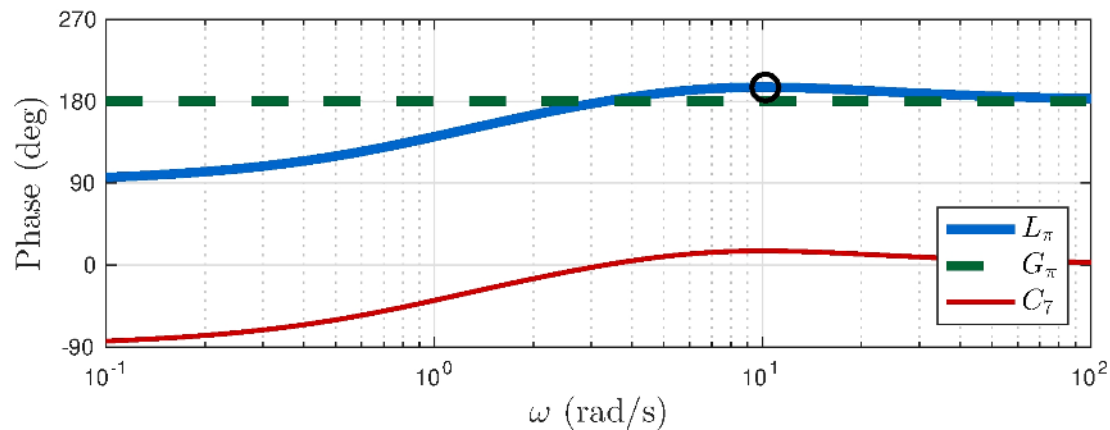
$$C(s) = K \frac{(s + z_1)(s + z_2)}{s(s + p_1)}$$

with

$$p_1 > z_2 \geq z_1$$

From C_7

$$\begin{aligned} p_1 &= 11, & K &= 3 \\ z_2 &= 5, & z_1 &= 1 \end{aligned}$$



Margins

$$\begin{aligned} G_M &\approx 0.55 \approx -5.2\text{dB}, & \phi_M &\approx 15^\circ \\ S_M &= 0.27 \end{aligned}$$

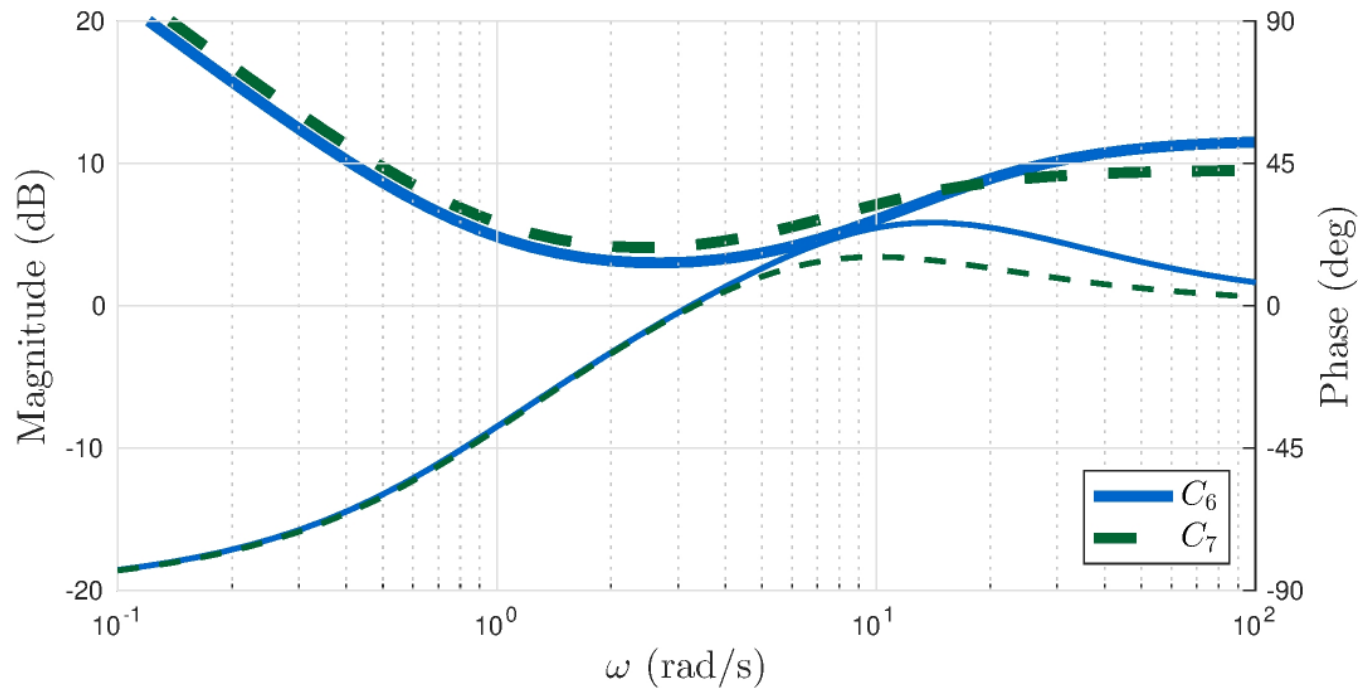
Control of the Simple Pendulum - Part II

Controller C_6

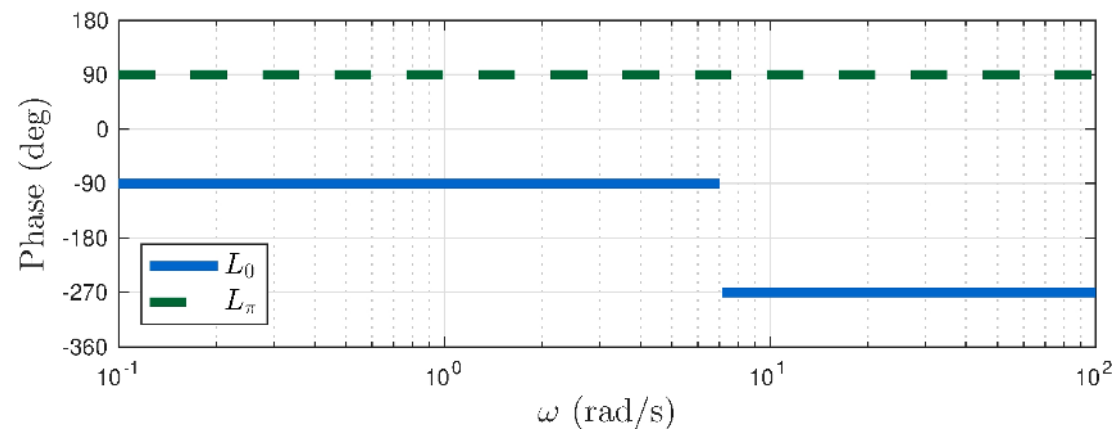
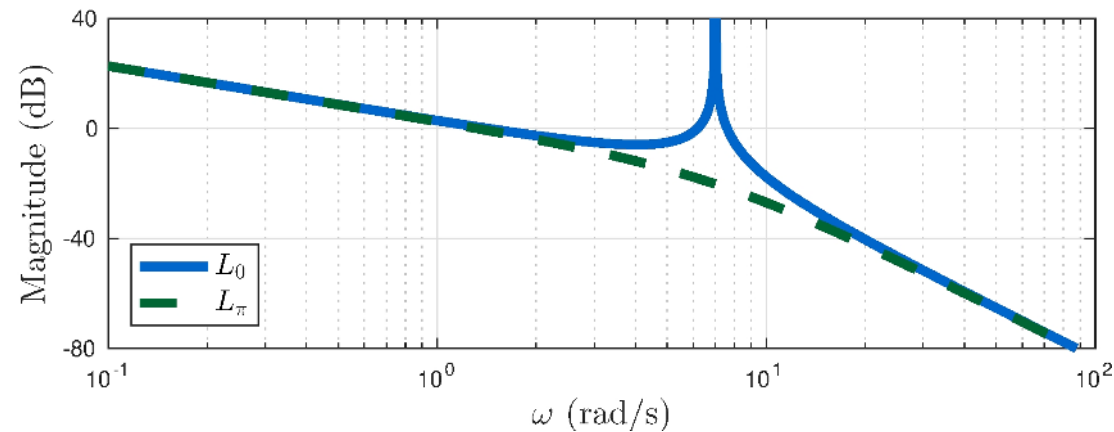
$$C_6(s) = 3.83 \frac{(s + 0.96)(s + 6.86)}{s(s + 21)}$$

Controller C_7

$$C_7(s) = 3 \frac{(s + 1)(s + 5)}{s(s + 11)}$$



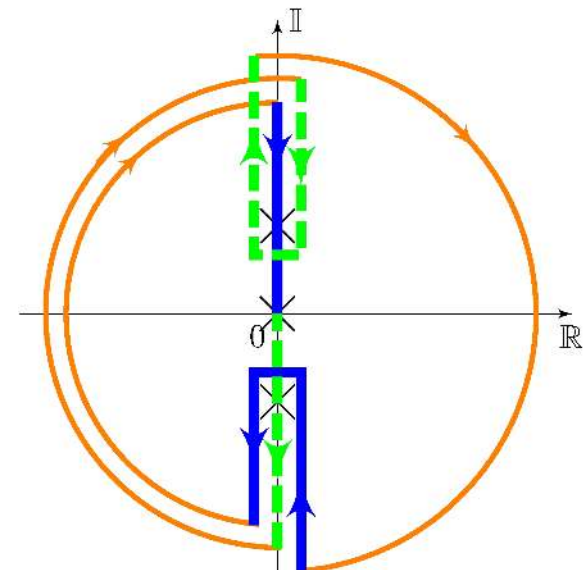
Control of the Simple Pendulum - Part II



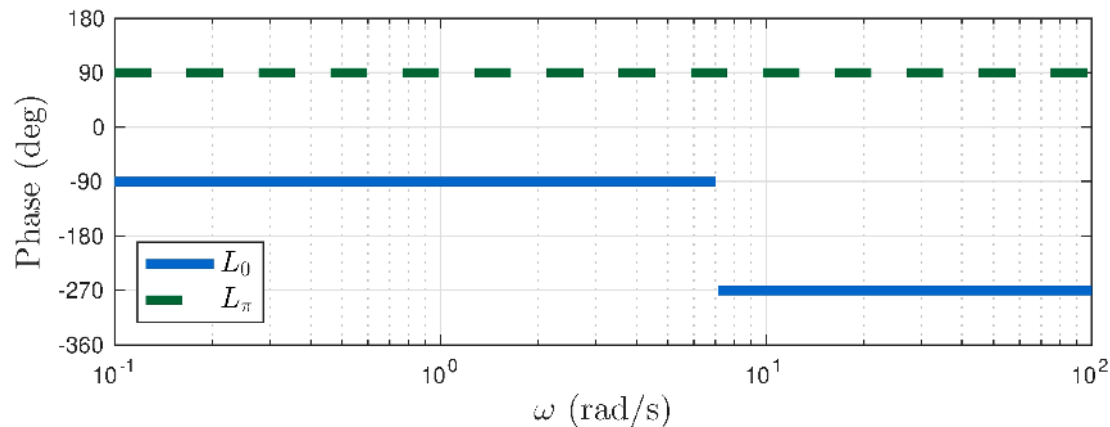
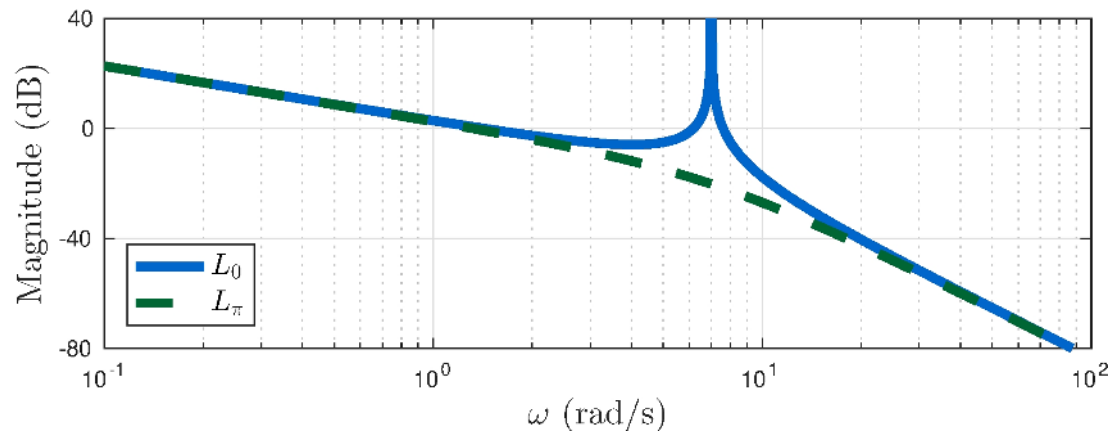
Nyquist analysis

$$L_0 = \frac{G_0}{s} \approx \frac{1.36}{s(1 + js/7)(1 - js/7)}$$

$$Z_\Gamma = P_\Gamma - \frac{1}{2\pi} \Delta_\Gamma^{-1} \arg L_\pi(s) \\ = 0 - (-2) = 2$$



Control of the Simple Pendulum - Part II



Stability analysis

Because $P_\Gamma = 0$, $L_0(j\omega)$ needs

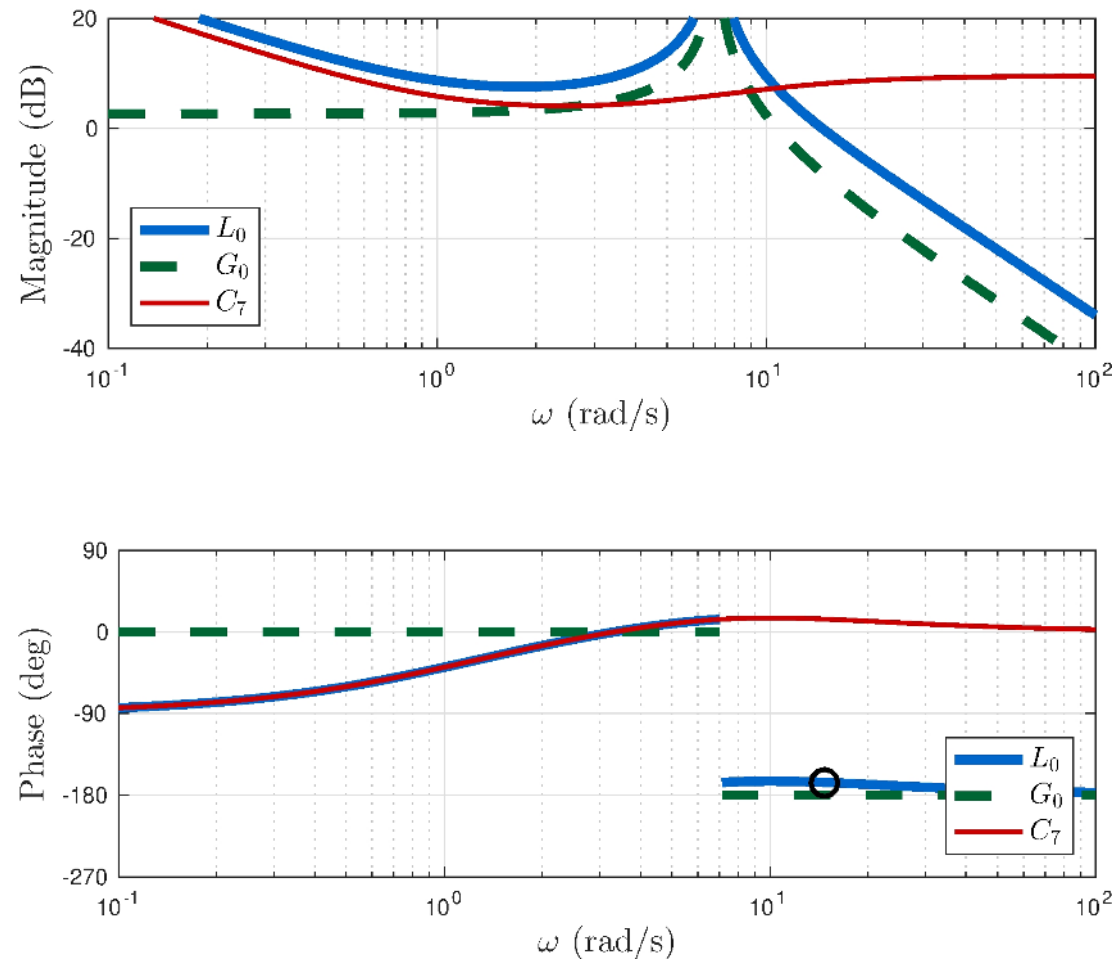
- no encirclements
- to not cross negative real axis

If more than 90° phase is added to $L_0(j\omega)$ there will be no crossing of the negative real axis nor encirclements. Controller must have

- at least two stable zeros
- at least one pole for realizability (in addition to the integrator)

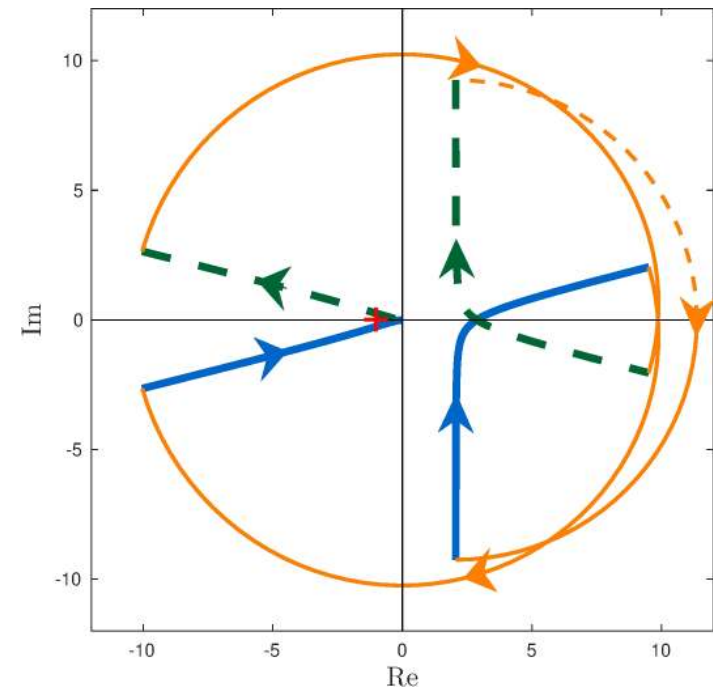
Same specs as for L_π !

Control of the Simple Pendulum - Part II



Controller C_7

$$C_7(s) = 3 \frac{(s+1)(s+5)}{s(s+11)}$$



Learn more

- [Book website](#)